



United International University

School of Science and Engineering

Final Examination; Year 2025; Trimester: Fall

Course: PHY 2105; Title: Physics; Sec: All

Full Marks: 40, Time: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Questions no 1, 2, 3,4,5,6,7 are mandatory to answer. Answer any one from question no 8 and 9.

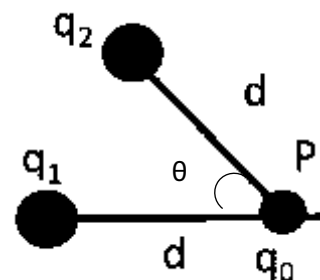
1. (a) The distance between two equal positive charges has been increased from r to $2r$. What will be the change in force between them? Discuss briefly from the perspective of Coulomb's Law. 2 CO1

- (b) Two points A and B are of equal potential V . How much work will be done in bringing a charge q from A to B? Discuss the reason using the formula of potential and work done. 2 CO1

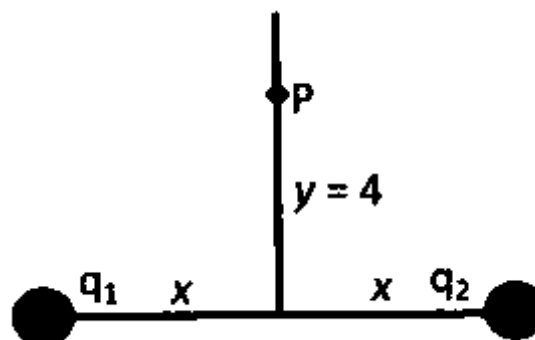
- (c) Ron found a charged ball with charge -3.2×10^{-15} C. If the charge of an electron is -1.6×10^{-19} C what will be the number of electrons in the ball? 2 CO1

2. Two charges $+5\mu\text{C}$ and $-10\mu\text{C}$ are placed 10 cm apart. What is the electrostatic force at the midpoint between the two charges if a positive charge of $+1\mu\text{C}$ is introduced there? 4 CO3

3. From the figure, $q_1 = 5$ C, $q_2 = 10$ C, $q_0 = 1$ nC, $d = 2$ m and the angle between q_1 and q_2 are $\theta = 60^\circ$. (i) Calculate x and y components of the net force on the test charge q_0 . 5 CO3
(ii) Calculate the magnitude and direction of the net force on q_0 .



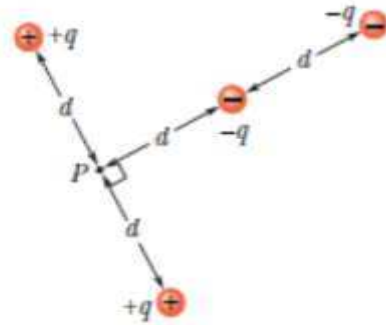
4. In figure, two charged particles on the x-axis: $q_1 = -10$ C is at $x = -3.00$ m and $q_2 = 5$ C is at $x = +3.00$ m. What are the (i) magnitude and (ii) direction of the net electric field produced at point P in the y-axis at $y = 4.00$ m. 5 CO3



5. The permanent dipole moment of $\vec{p} = (5\hat{i} + 2\hat{j}) \times 10^{-30}$ Cm is placed in an external electric field $\vec{E} = 5000\hat{i} + 6000\hat{j}$ N/C. 6 CO3
(i) Calculate the torque experienced by the dipole (magnitude with direction).
(ii) What is the potential energy of the dipole in that field?

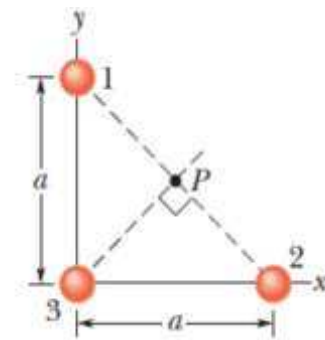
6. In the adjacent figure, what is the net electric potential at point P due to the four particles if $V = 0$ at infinity, $q = 6$ nC, and $d = 2$ m?

3 CO3



7. The three particles are fixed in place and have charges $q_1 = q_2 = +2e$ and $q_3 = +3e$. Distance $a = 5$ μm . What is the magnitude of the net electric potential at point P due to the particles?

3 CO3



8. (a) Show that electric field on the axis of an electric dipole is, $E = \frac{1}{2\pi\epsilon_0} \cdot \frac{qd}{z^3}$. Here the symbols have their usual meanings. 4 CO2

- (b) Show that, the torque experienced by a dipole in an electric field is, $\vec{\tau} = \vec{P} \times \vec{E}$, where $\vec{P}$ is the dipole moment and $\vec{E}$ is the electric field. Calculate the condition for maximum and zero torque. 4 CO2

9. (a) Show that, the electric potential difference between two points i and f is 4 CO2

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}.$$

where, V , E and ds denote potential, electric field and length element, respectively.

- (b) Show that, the electric potential due to a point charge at a distance r from it is, 4 CO2

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}.$$

where, the symbols have their usual meanings.

CO1: Define different physical quantities with examples. **CO2:** Derive/Show the various equations of electric field, electric potential, electric dipole, dipole moment, electrostatic force, etc. **CO3:** Evaluate different numerical problems based on the basic characteristics of electric charge, electric field, electric potential, electric dipole moment, and electrostatic force, etc.